

Olist E-Commerce End-to-End Analytics Project

Objective

Designed and implemented a production-grade end-to-end data analytics pipeline on the Brazilian Olist e-commerce dataset. Progressed raw transactional data through a validated Medallion Architecture into a seven-dashboard business intelligence suite and a 36-page analytical report with quantified strategic recommendations.

Tools Used

- Databricks
- Python (PySpark)
- SQL
- Tableau

Dataset

Brazilian Olist Marketplace (~100k orders)

Deliverables

- Data pipeline (Bronze, Silver, and Gold Medallion layers)
- SQL query layer linking the validated gold layer to all Tableau visualisations, with full traceability between pipeline output and dashboard metrics
- 7 Tableau dashboards highlighting key trends and insights at the product, regional, seller, and customer levels
- A 36-page BI report detailing key data insights and strategies

Olist Sales Performance & Fulfillment Analysis

Key metrics and trends across revenue, operations, and logistics

Total Orders

96.47K

Total Revenue

R\$15.42M

Avg Order Value

R\$198.54

Avg Delivery Time

18.7 Days

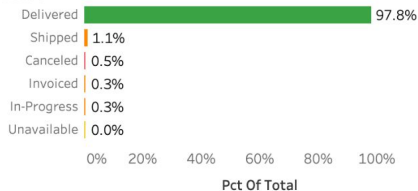
Avg Review Score

4.1

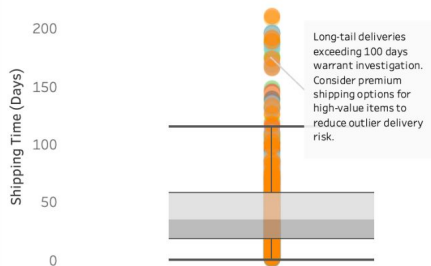
Order Status Funnel

Nearly all customer orders are successfully delivered, demonstrating efficient fulfillment. Order cancellations and unavailable products remain minimal, suggesting healthy inventory management.

Order Status



Order Delivery Distribution by Product Category



Monthly Sales Revenue 2017-2018 for Top 7 Product Categories

Revenue experienced rapid growth throughout 2017 before stabilizing in 2018, suggesting successful platform expansion followed by market maturity.

Living & Home Appliances, Sports, Toys & Leisure, and Electronics and Gaming account for almost half of all platform sales, indicating strong dependence on a small number of product categories.

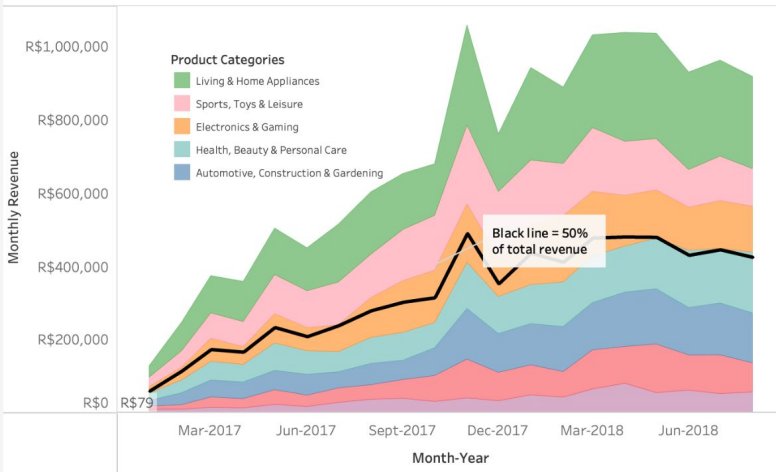
Suggest:

Invest marketing in top-performing categories

Diversify revenue by promoting weaker categories

Filter: Product Category

(Multiple values)



Business Insights

- Processed 96K+ orders generating R\$15.4M GMV with a 97.8% delivery success rate, demonstrating strong operational execution.
- Revenue accelerated throughout 2017 before stabilizing in 2018, signaling marketplace maturation rather than slowing demand.
- 3 categories account for ~50% of total revenue, leaving platform performance disproportionately sensitive to demand shocks in these segments.
- Long-tail deliveries exceeding 100 days were identified across multiple categories, pointing to fulfillment bottlenecks.

Strategic Recommendations

- Diversify revenue mix by increasing marketing investment in high-growth, freight-efficient categories (e.g., Health, Beauty & Personal Care)
- Introduce premium shipping for high-value products to reduce long-tail delivery delays.
- Monitor the relationship between seller count and orders-per-seller monthly.

[Link to Tableau Public Interactive Dashboard](#)

Freight Burden Analysis

Analysis restricted to products with total order count ≥ 5

Products Analyzed

4,189

Pipeline Risk Products

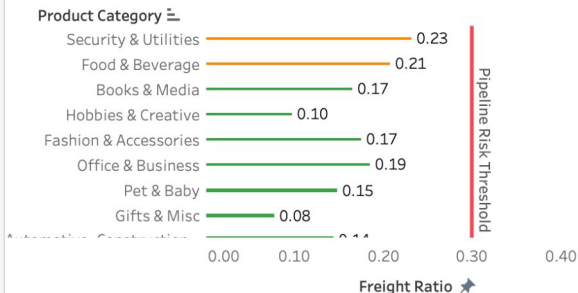
904

Revenue At Risk*

R\$540,984

*From products with freight ratio $\geq 30\%$

Freight Burden by Product Category



Products above the 30% freight ratio threshold (pipeline risks) are candidates for repricing, supplier renegotiation, or delisting

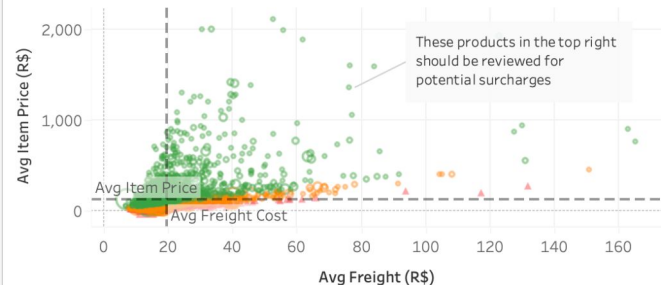
▲ The scatterplot cluster below R\$50 avg price shows the highest density of pipeline risk products. Low-price items cannot absorb high freight costs and are the most margin-vulnerable.

▲ Products in the top-right quadrant (high price, high freight) are candidates for freight surcharges rather than delisting.

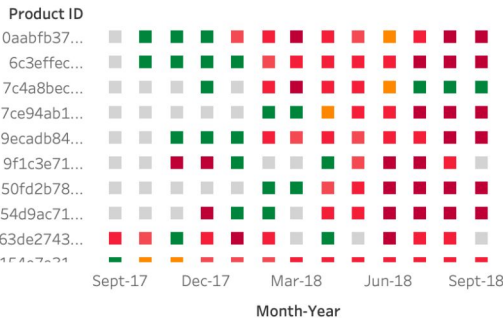
Action: The table below is filtered to pipeline risk products with multiple 3-month growth cycles of $\geq 40\%$ or less. Review for specific SKUs flagged across multiple consecutive months of decline.

Identifying Products Where Shipping Costs Erode Margin

Products above the 30% freight ratio threshold are candidates for repricing, supplier renegotiation, or delisting.



Pipeline Risk Products: Review for Action



Business Insights

- 904 products (21.6% of the analysed catalog) exceed the platform's 30% freight-risk threshold, placing R\$541K of revenue under margin pressure.
- Products below R\$50 in average selling price face the most acute freight burden.
- 24 products satisfy a two-way risk criterion simultaneously: freight ratio $\geq 30\%$ and rolling 3-month demand decline $\geq 40\%$ for multiple cycles. These represent the highest-urgency commercial action items on the platform.

Strategic Recommendations

- Initiate immediate commercial review of the 24 dual-flagged products and delist those with no path to margin recovery.
- Introduce a freight surcharge policy for high-price, high-freight products where the margin exists to absorb a price adjustment without losing the customer.
- Expand investment in Health, Beauty & Personal Care as a structurally advantaged growth category with favourable shipping economics.

[Link to Tableau Public Interactive Dashboard](#)

Freight Burden Flag

3-Month Growth %

Freight Ratio Band

○ Healthy
▲ Pipeline Risk

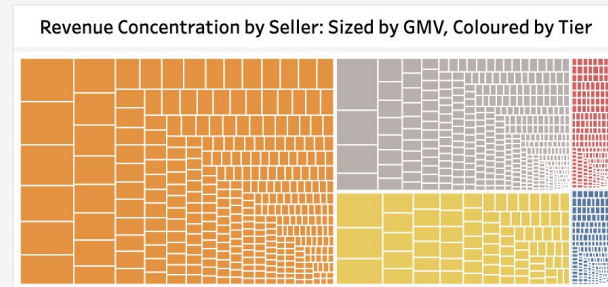
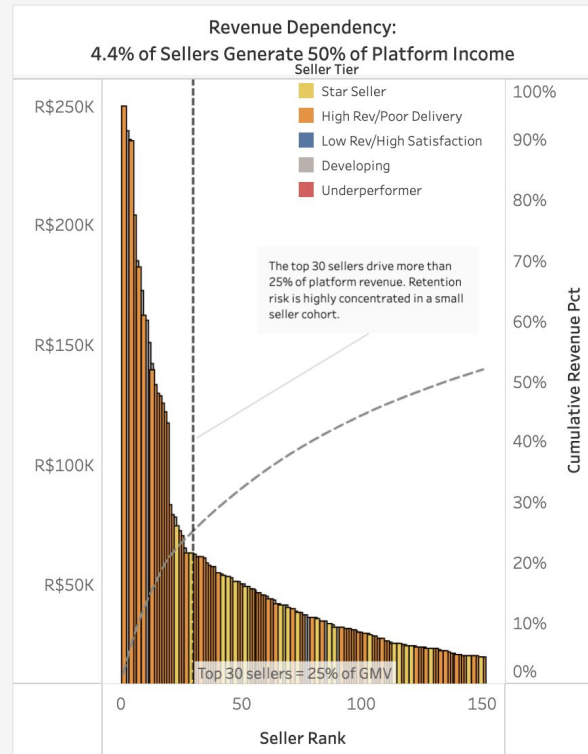
Filter: Freight Burden

(All)

Seller Revenue Concentration and Platform Growth:

Where Risk and Opportunity Are Concentrated

Top 136 sellers generate ~50% of platform GMV. Retention of this cohort is the single largest revenue risk on the platform. Active seller count and orders per seller both grew through 2017–2018, signalling healthy marketplace expansion.



High Rev / Poor Delivery sellers (orange) dominate the top revenue blocks, accounting for the largest share of platform GMV. Despite delivery delays, their review scores suggest customers value their products. However, delivery performance remains a latent churn risk.



Business Insights

- 50% of revenue is generated by only 136 sellers (4.4%); this creates significant revenue concentration risk.
- High Revenue / Poor Delivery sellers generate 48% of GMV while averaging delivery times 27% slower than Star Sellers. Their satisfaction scores sit at the platform average, suggesting customers are tolerating delays due to product quality.
- Low Revenue / High Satisfaction sellers demonstrate solid operational performance but are commercially underutilized.

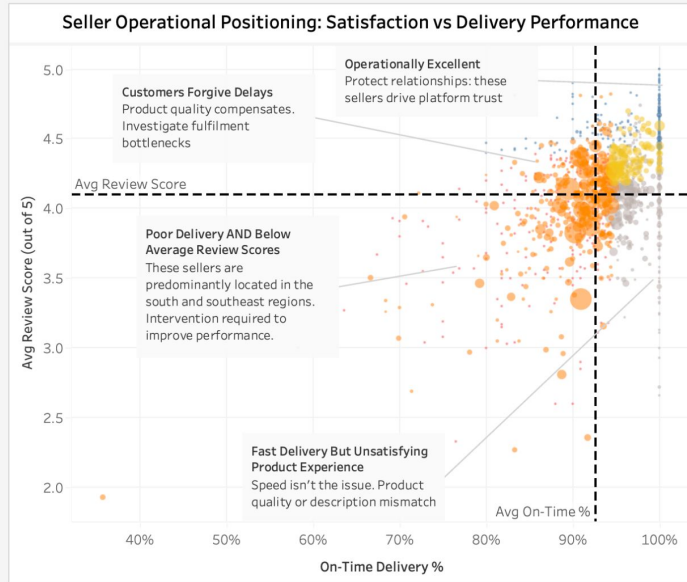
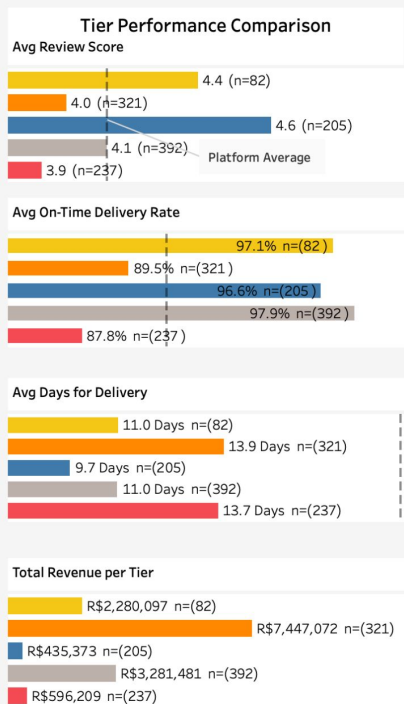
Strategic Recommendations

- Establish a formal retention programme for the top 136 sellers with dedicated account management, early contract renewal incentives, and priority dispute resolution.
- Engage High Revenue / Poor Delivery sellers with proactive SLA targets and logistical support before satisfaction scores deteriorate. A 5% improvement in this tier's on-time rate would materially reduce platform-wide late delivery complaints.
- Invest marketing and catalog support into high-performing emerging sellers to reduce revenue concentration.

[Link to Tableau Public Interactive Dashboard](#)

Seller Performance Scorecard: Operational Health and Revenue by Tier

3,095 active sellers (with >=10 sales each) ranked across delivery speed, customer satisfaction, and revenue contribution. High Rev / Poor Delivery sellers represent the platform's most urgent operational risk: significant gross merchandise value at risk from a degraded customer experience.



STRATEGIC PRIORITY SUMMARY

- Immediate:** Engage High Rev/Poor Delivery sellers (48% of GMV, 13.9 avg delivery days) with SLA improvement targets. A 5pp on-time improvement in this tier materially reduces late delivery complaints across the platform.
- Short-Term:** Identify Low Rev/High Satisfaction sellers with capacity to scale. Their operational metrics already meet Star Seller standards: commercial support is the missing ingredient.
- Monitor:** Underperformer tier. Consider minimum performance thresholds and probationary framework for sellers below platform averages on both metrics simultaneously.

Business Insights

• Sellers were scored across revenue contribution, on-time delivery rate, and customer satisfaction and then assigned to one of five tiers using NTILE quartile scoring:

- **Star Sellers** set the operational benchmark: 97.1% on-time delivery, 4.4 average review score, and 11-day average delivery time. These 82 sellers define excellence.
- **The High Revenue / Poor Delivery tier** is the most strategically complex: they generate 48% of GMV but average 13.9 delivery days and hold the lowest review scores among commercially significant tiers. Customers are currently forgiving delays, but this reflects a lagging indicator.
- **Underperformers** score below platform average on both delivery and satisfaction. At R\$596K total revenue the individual stakes are low, but cumulative poor experience across this tier has a measurable drag on platform-wide satisfaction scores.

Strategic Recommendations

- **Immediate:** Engage High Revenue / Poor Delivery sellers with SLA improvement targets and logistical support before satisfaction scores further deteriorate.
- **Short-term:** Identify Low Revenue / High Satisfaction sellers with catalogue depth and capacity to scale. Marketing visibility and commercial support are their barriers to migration.
- **Monitor:** Introduce minimum performance thresholds for the Underperformer tier with a defined improvement window and clear consequences for non-compliance.

[Link to Tableau Public Interactive Dashboard](#)

Regional Sales Landscape

Comparing market size, revenue growth, seller density, and delivery performance across Brazilian states

Total Revenue

R\$15.42M

Total Revenue from Southeast

R\$9.96M

Highest Revenue State: São Paulo

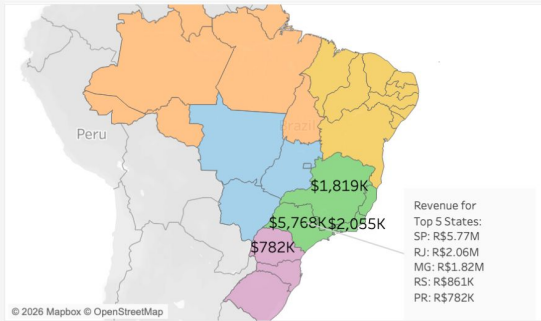
R\$5.77M

Largest Delivery Gap State: Alagoas

-14.6% to National Avg

Revenue Concentration by State

Southeast Brazil accounts for the majority of platform revenue



Customer Region

- Central-West
- North
- Northeast
- South
- Southeast

Filter: Region

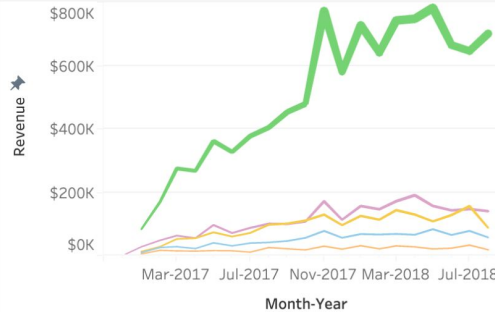
(All)

Filter: State

(All)

All regions reached highest sales between Apr-Jul 2018

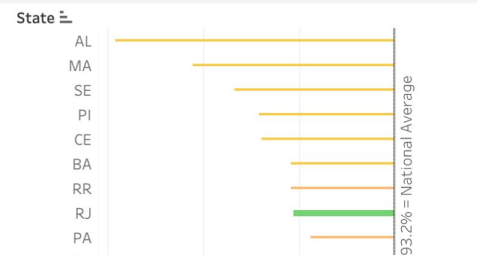
Southeast consistently generated ~65% of total revenue throughout 2018



Seller Coverage vs Revenue by State



States Delivering Faster (or Slower) Than Average



Business Insights

- The Southeast generates 64.6% of all revenue, with São Paulo alone contributing 37.4%, highlighting heavy geographic concentration.
- Northern and Northeastern states consistently underperform in delivery speed, suppressing customer satisfaction and future demand.
- Bahia represents a high-potential expansion opportunity where demand currently exceeds local seller coverage.

Strategic Recommendations

- Launch a targeted seller recruitment campaign in Bahia, where revenue demand currently exceeds local seller supply.
- Prioritise seller recruitment across Northern and Northeastern states to address the structural delivery performance gap.
- The Southeast is approaching saturation. The Northeast and South offer the best combination of population density and untapped demand for the next growth phase.

[Link to Tableau Public Interactive Dashboard](#)

Customer Sentiment and Value Segmentation: Understanding Who Is Satisfied, Who Is At Risk, and What It Costs

95,026 customers segmented by lifetime spend and satisfaction score. R\$1.1M in high-value customer revenue sits with dissatisfied buyers, representing the platform's most urgent retention blind spot.

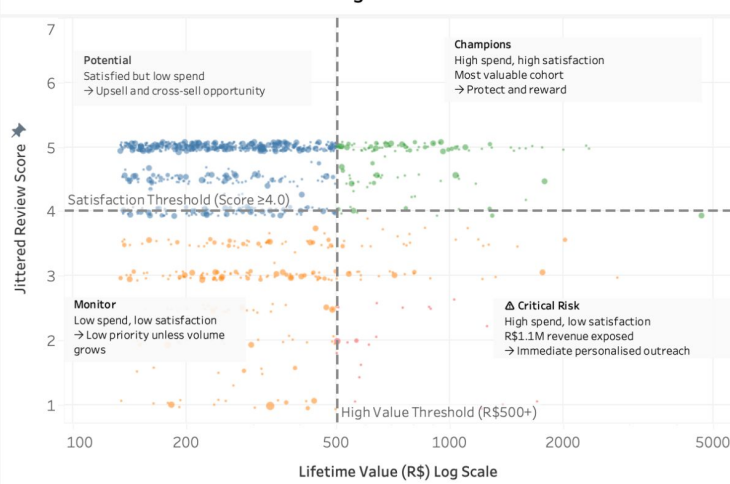
Revenue and Customer Count by
Value-Sentiment Quadrant

Value Tier	Sentiment Tier	
	At Risk / Dissatisfied (<4.0)	Satisfied (≥4.0)
High Value	R\$1,141,465 1,226 Customers (1.3%) R\$931 Avg/Customer	R\$2,900,320 3,157 Customers (3.3%) R\$919 Avg/Customer
Low Value	R\$2,812,815 20,544 Customers (21.6%) R\$137 Avg/Customer	R\$8,882,924 70,099 Customers (73.8%) R\$127 Avg/Customer

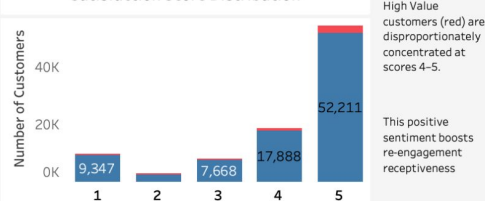
The High Value / At Risk quadrant contains only 1.3% of customers but represents R\$1.1M in exposed revenue. These customers have spending patterns consistent with Champions but satisfaction scores that signal a negative experience: most likely a late delivery or product quality issue. Their average spend of R\$931 per customer makes individualised outreach economically justifiable.

The Low Value / Satisfied quadrant is the platform's largest cohort by volume. High satisfaction suggests these customers are open to repeat engagement. A loyalty incentive programme targeting this segment has the highest addressable market of any quadrant.

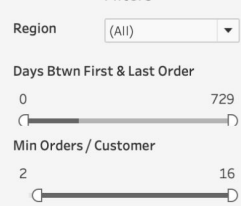
Individual Customer Positioning: Lifetime Value vs. Review Score



Satisfaction Score Distribution



Filters



Business Insights

- 1,226 high-value customers account for R\$1.14M and represent the most valuable recovery opportunity.

- High-value customers spend nearly 7 times more than low-value customers, justifying personalized retention investment.

- Most customers remain satisfied, but dissatisfaction among high-value buyers carries disproportionate financial risk.

Strategic Recommendations

- Launch personalized recovery campaigns targeting dissatisfied high-value customers.

- Introduce a loyalty programme targeting the Low Value / Satisfied customers. Their satisfaction signals openness to re-engagement, and even a modest increase in purchase frequency across this cohort would generate meaningful incremental revenue.

- Investigate the root causes of one-star reviews systematically. At almost 10K customers, this segment is large enough to be a structural platform quality issue rather than random negative experiences.

[Link to Tableau Public Interactive Dashboard](#)

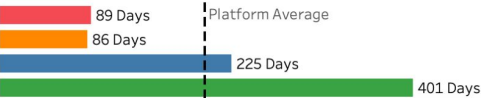
Customer Reactivation Priority: Where to Focus Re-engagement Efforts

Reactivation Tiers:

Customer Satisfaction & Delivery Performance

R\$8.1M in lifetime value is held by almost 39k inactive customers across 4 priority tiers. Tier 1 (Urgent): High value customers silent <90 days represent the highest ROI reactivation target.

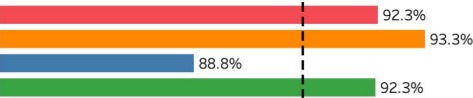
Avg Days Since Last Order



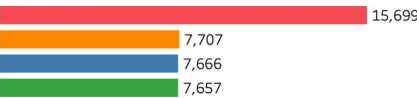
Avg Satisfaction Score (out of 5)



Avg On-Time Delivery %



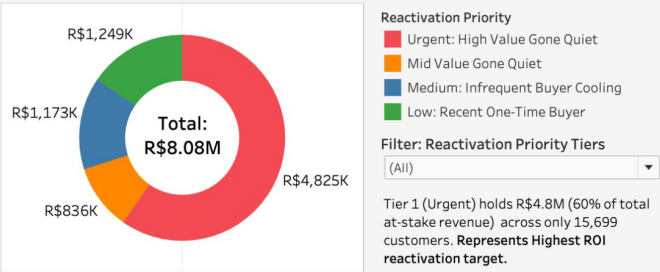
Customer Count



Churn appears situational, not experiential

Tier 1 customers have been silent for an average of 88.8 days but retain high satisfaction (4.1) and strong delivery experience (92.3% on time).

Revenue At Stake



Reactivation Priority

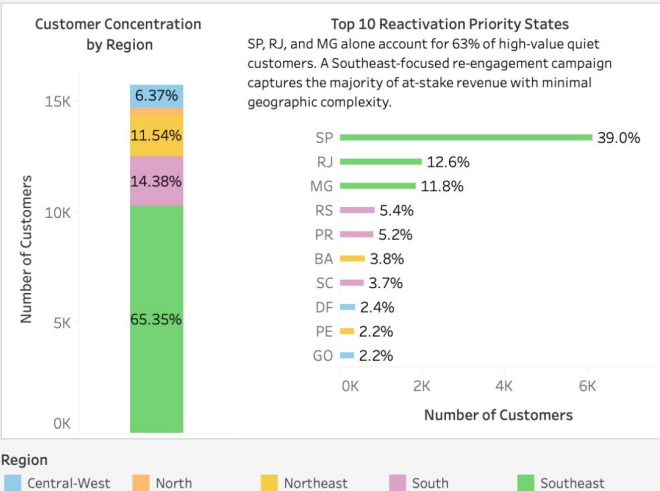
- Urgent: High Value Gone Quiet
- Mid Value Gone Quiet
- Medium: Infrequent Buyer Cooling
- Low: Recent One-Time Buyer

Filter: Reactivation Priority Tiers

(All)

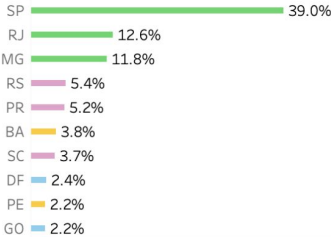
Tier 1 (Urgent) holds R\$4.8M (60% of total at-stake revenue) across only 15,699 customers. Represents Highest ROI reactivation target.

High Value/Quiet Customer Base Target Regions



Top 10 Reactivation Priority States

SP, RJ, and MG alone account for 63% of high-value quiet customers. A Southeast-focused re-engagement campaign captures the majority of at-stake revenue with minimal geographic complexity.



Region

Central-West North Northeast South Southeast

Business Insights

• R\$8.08M in lifetime value is held by 38,729 inactive customers across 4 reactivation priority tiers: recoverable revenue that requires re-engagement, not acquisition spend.

• 15,699 inactive, high-value customers (Tier 1) represent R\$4.8M (over half of at-stake revenue) while maintaining positive satisfaction scores, indicating low-friction recovery potential.

• SP, RJ, and MG states alone account for 63% of high-value quiet customers. A Southeast-focused campaign captures the majority of at-stake revenue with minimal geographic complexity.

Strategic Recommendations

• Launch Tier 1 re-engagement campaigns immediately, before the 89-day average silence window extends further. Tier 1's High satisfaction and recent purchase history mean campaigns face low friction.

• Concentrate initial campaign geography on SP, RJ, and MG to capture 63% of at-stake revenue at the lowest operational complexity. Expand to secondary states only after validating initial conversion rates.

[Link to Tableau Public Interactive Dashboard](#)